



10 : 04 : 22 : 33
DAY HRS MIN SEC

Gridlock Hackathon 2.0

LIVE

May 26, 2026, 10:00 PM IST (Asia/Kolkata) - Jun 07, 2026, 11:59 PM IST (Asia/Kolkata)

INSTRUCTIONS

PROBLEMS

SUBMISSIONS

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[← Problems](#) / Traffic demand prediction

Traffic demand prediction

Max. score: 100

Cities worldwide are increasingly turning to AI-powered solutions to tackle the issue of traffic congestion. This problem disrupts the smooth flow of transportation and poses a significant barrier to economic growth. To address this challenge effectively, the first step is to understand travel demand and patterns within urban areas comprehensively. By harnessing the power of AI, cities, and regions aim to gather critical insights into transportation dynamics. This will enable them to implement data-driven strategies and solutions to alleviate traffic congestion and promote more efficient mobility. Ultimately, this endeavor will foster economic development and prosperity.

Task

Design a system that helps us provide valuable insights into passenger travel patterns, booking behavior, and trip cancellations, which can be used for various analyses and predict demand in the travel industry.

Dataset description

The dataset folder contains the following files:

- **train.csv:** 77299 x 11
- **test.csv:** 41778 x 10
- **sample_submission.csv:** 5 x 2

Variable description

The columns provided in the dataset are as follows:

Column name	Description
Index	Represents the unique identification of datapoint
geohash	Represents geographic information regarding a place
day	Represents the day when the information is recorded
timestamp	Represents the timestamp of the record inserted in the system

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RoadType	Represents the type of road in the nearby location
NumberofLanes	Represents the number of roads/lanes present in the location
LargeVehicles	Represents whether large vehicles are permitted on the specific roads/lanes
Landmarks	Represents whether there are any landmarks near the location
Temperature	Represents the temperature of the place
Weather	Represents the weather of the place
demand	Represents the demand of the traffic at the timestamp

Evaluation metric

```
score = max(0,100*(metrics.r2_score(actual,predicted)))
```

Result submission guidelines

- The index is "Index" and the target is the "demand" column.
- The submission file must be submitted in .csv format only.
- The size of this submission file must be 41778 x 2.
- Ensure that your submission file contains the following:
 - Correct index values as per the test file
 - Correct names of columns as provided in the *sample_submission.csv* file

Instructions

1. Click *Download dataset* to download the dataset.
2. Solve the problem in your local environment.
3. Save the predictions in a .csv file.
4. Click *Upload File* (under the *Upload File* section) to upload your prediction file (.csv).
5. Click *Upload File* (under the *Upload Source Code* section) to upload your .ipynb file along with any presentation file.
6. Add any instructions or comments in the *Your Answer* section.
7. Click *Submit*.

[Download dataset](#)

Upload Prediction File

Please upload the prediction file in the format as stated in the problem.

Choose File No file chosen

Submit & Evaluate

Upload Source Files

You need to submit a zip or tar archive consisting of a text file explaining your approach, details about feature engineering, tools you used and the relevant source files.

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Choose File

No file chosen

Upload

COMMENTS (27) SORT BY: **Relevance** ▼

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**Yagyansh Dev Singh** a day ago

Just wrapped up a traffic demand forecasting challenge — predicting normalized ride demand across locations and time slots using geohash, road, and weather data.

The biggest unlock was geohash × time-slot target encoding — mapping each location's historical demand at every 15-minute window. This single interaction feature captured rush-hour patterns specific to each location far better than any raw feature could.

Other key moves:

Cyclical sin/cos encoding for time (so midnight doesn't look far from 11:45 PM)

Approximate lat/lon decoded from geohash strings for continuous spatial signal

Temperature imputed via group medians (geohash prefix + day) before global fallback

LightGBM with 2000 trees, leaf-wise growth, and light L1/L2 regularization

Final R^2 of 0.9994 on the training set.

The lesson: in spatio-temporal problems, interaction encodings between location and time almost always matter more than engineering either dimension alone.

▲ 10 votes ● Reply

**Shraddha Rajendra Jadhav** 20 hours ago

I want to join ur team.I have technical expertise in AIML, Fullstack

Development. I was

Shortlisted, Selected in

Nationals for Amazon,

Google, SIH, etc

Hackathons. My Mail id-

sjn202358@gmail.com

▲ 0 votes ● Reply

**Ayush Ranjan** 20 hours ago

We are looking for a teammate, we are experienced in competitions like this, having won two bronze medals at kaggle challenges. If anyone is up please reply to this comment !

▲ 2 votes ● Reply

**Aditya Chaudhary** 19 hours ago

theaditya1985@gmail.com

▲ 0 votes ● Reply

**Lokesh**  Edited 3 hours ago

quan655737@gmail.com , deeply knows about kaggle playground competitions(iykyk)

▲ 0 votes ● Reply

**Shreyash Pore** a day ago

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Building AI-driven traffic demand prediction to transform urban mobility with smarter, data-backed insights. Our solution analyzes travel patterns, road conditions, weather, and peak-hour trends to predict congestion accurately.
Focused on reducing traffic chaos, improving city transportation efficiency, and enabling smarter future-ready infrastructure

▲ 0 votes ● Reply



Shraddha Rajendra Jadhav 20 hours ago

would like to join ur team.I have technical expertise in AIML, Fullstack Development. I was Shortlisted, Selected in Nationals for Amazon, Google, SIH, etc Hackathons. Mail id- sjn202358@gmail.com

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Shib Chandan Mistry a day ago

Approach Summary:

1. Preprocessing: Merged datasets to ensure aligned categorical transformations between train and test distributions.
2. Feature Engineering: Extracted structural information from geohash coordinates by slicing text into hierarchical prefixes to capture regional dynamics. Transformed timestamps using sine and cosine functions to model daily cyclical peak hours.
3. Modeling: Deployed a 5-fold cross-validation routine coupled with a gradient boosting architecture to minimize validation error and prevent overfitting.
4. Output: Generated predictions aligned with target evaluation rules.

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Shraddha Rajendra Jadhav 20 hours ago

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Monalisha Patra a day ago

- Developed a leakage-free traffic demand prediction pipeline using LightGBM and advanced feature engineering techniques.
- Engineered temporal, spatial, weather, and infrastructure-based features to effectively model traffic behavior patterns.
- Implemented hierarchical geohash-based target encoding with fallback mechanisms to handle sparse and unseen regions.
- Identified and resolved a critical target leakage issue that was artificially inflating validation performance.
- Applied fold-wise target encoding to ensure reliable cross-validation and robust model generalization.
- Utilized Bayesian smoothing and ensemble modeling to improve prediction stability and reduce overfitting.
- Achieved a stable and trustworthy out-of-fold score of approximately 93.32 through leakage-free evaluation.

▲ 1 vote ● Reply



Shraddha Rajendra Jadhav 20 hours ago

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Chetan Jain a day ago

LightGBM with day-48 demand lookup as the golden feature. Key insight: all test timestamps (Day 49) exist in Day 48 training data, so demand_d48 (same geohash + timestamp from Day 48) was the strongest predictor. Added geohash-decoded lat/lon, per-location & per-timestamp aggregate stats, cyclical time encoding, and weather/road features. 5-fold CV score: 98.31 $R^2 \times 100$

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Shraddha Rajendra Jadhav 20 hours ago

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Magraj Jakhar a day ago

Two-model ensemble (LightGBM + XGBoost) with spatial-temporal feature engineering and Day 48 lag features. Local validation $R^2 = 94.13$. Final prediction is a weighted blend of a global model (20%) and a lag-aware model (80%).

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Shraddha Rajendra Jadhav 20 hours ago

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Abneesh Singh a day ago

Solution Overview: Spatial-Temporal Forecasting Ensemble
Architecture:

We engineered a 63-feature pipeline focused on hyper-local, time-series traffic patterns. Key features include label-encoded geohashes (allowing gradient boosting trees to learn location-specific baseline signatures) and temporal interactions (e.g., RoadType $\times$ Hour). We extracted extensive baseline statistics for each (Geohash + Hour) combination, as our variance decomposition proved this explains 94.8% of historical traffic variance.

Modeling Strategy:

Base Models: LightGBM, CatBoost, and XGBoost regression models.

Hyperparameter Tuning: Optuna was used to run hundreds of optimization trials to mathematically determine optimal depth, learning rate, and regularization settings via cross-validation.

Final Ensemble: Predictions were generated using a weighted blend of the base models, with weights optimized via a secondary Optuna study to minimize RMSE and maximize the R^2 score.

Generalization: The final model was trained on the entire dataset to maximize robustness against di ? on shifts in the daytime test predictions.

 0 votes  Reply**Shraddha Rajendra Jadhav** 20 hours ago

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 0 votes  Reply**23ucc555** 20 hours ago

hello

 0 votes  Reply**Sakthi J** 20 hours ago

Built a traffic demand prediction model using XGBoost with feature engineering on timestamp and categorical encoding. Used preprocessing, label encoding, and regression modeling to predict traffic demand efficiently.

 0 votes  Reply**Shraddha Rajendra Jadhav** 20 hours ago

would like to join ur team.I have technical expertise in AIML, Fullstack Development. I was Shortlisted, Selected in Nationals for Amazon, Google, SIH, etc Hackathons. Mail id- sjn202358@gmail.com

 0 votes  Reply**SINGAVARAM HEERALAL SINGH** 13 hours ago

Just wrapped up a Traffic Demand Prediction project using Machine Learning ??

- ✓ Performed data preprocessing and feature engineering
- ✓ Extracted time-based insights from timestamp data
- ✓ Handled categorical and numerical features efficiently
- ✓ Built a Random Forest Regression model for prediction
- ✓ Generated prediction results in required CSV format

Tech Stack Used:

Python | Pandas | Scikit-learn | Machine Learning

This project helped me understand how AI can be used for smart traffic management and urban mobility analysis. ? ✨

#MachineLearning #Python #DataScience #AI #TrafficPrediction #MCA #ScikitLearn #Analytics

 0 votes  Reply**Kishan Kumar** 7 hours ago

Very Good Problem Statement

 0 votes  Reply**utkarshv087** 7 hours ago

Built a traffic demand prediction model using feature engineering on timestamp and categorical encoding. Used preprocessing, label encoding, and regression modeling to predict traffic demand efficiently

 0 votes  Reply**YASASRI PRASANNA Y** 7 hours ago

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5/28/26, 7:36 PM

Traffic demand prediction - Gridlock Hackathon 2.0 | HackerEarth

Completed the Traffic Demand Prediction challenge in Flipkart Gridlock Hackathon 2.0 ?
Worked on preprocessing, feature encoding, and machine learning model training using Python and Random Forest.
Great learning experience in real-world ML problem solving ?
▲ 0 votes ● Reply



Ritesh Kumar Singh7 hours ago

Implemented an advanced ensemble-based traffic demand prediction pipeline using LightGBM, XGBoost, and CatBoost with Optuna hyperparameter tuning, target encoding, cyclic temporal feature engineering, and pseudo labeling for robust generalization and improved R² performance.
▲ 0 votes ● Reply



Loksai Palyam5 hours ago

Building a pure-Rust urban mobility AI engine that bypasses heavy Python environments. Decodes alphanumeric geographic geohashes into real-world coordinate vectors (Lat/Lon). Trains a high-fidelity 300-tree Random Forest to predict traffic demand loops.
▲ 0 votes ● Reply



Riya Gupta4 minutes ago

are we going to have another problem statement coming up?
▲ 0 votes ● Reply

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